MODULE 1: CHEMICAL THERMODYNAMICS AND KINETICS (CO: 1)

Complete Comprehensive Textbook: Laws of Thermodynamics, Carnot Cycle, Gibbs-Helmholtz, Rate Laws & Solved VIT FAT Question Bank

Aligned with Official VIT BACHY105 Applied Chemistry Course Syllabus, Department Question Bank & VIT FAT Papers

Syllabus Scope & Competency Checklist (Module 1):

- **Thermodynamics Terminologies:** System (open, closed, isolated), surroundings, boundary; state functions (U, H, S, G) vs path functions (w, q); extensive vs intensive properties; reversible vs irreversible processes.
- **Zeroth & First Laws of Thermodynamics:** Thermal equilibrium; mathematical formulation of First Law ($dU = dq + dw$); internal energy, expansion work ($w = -\int P_{ext} dV$); enthalpy ($H = U + PV$); heat capacities ($C_v = \left(\partial \frac{U}{\partial} T\right)_v, C_p = \left(\partial \frac{H}{\partial} T\right)_p$); Mayer's relation ($C_p - C_v = R$).
- **Gas Expansions:** Reversible and irreversible isothermal expansion of ideal gas ($w_{rev} = -nRT \ln\left(\frac{V_2}{V_1}\right)$); adiabatic expansion ($dq = 0, PV^\gamma = \text{const}, TV^{\gamma-1} = \text{const}, w = nC_v(T_2 - T_1)$).
- **Second Law of Thermodynamics & Carnot Engine:** Limitations of First Law; Kelvin-Planck and Clausius statements; Carnot cycle 4-stage operation (isothermal expansion, adiabatic expansion, isothermal compression, adiabatic compression); Carnot efficiency ($\eta = \frac{T_H - T_C}{T_H} = \frac{w_{net}}{q_H}$); Carnot theorem.
- **Entropy, Free Energies & Third Law:** Clausius inequality ($\oint d\frac{q_{rev}}{T} = 0$); entropy changes ($\Delta S$); Helmholtz energy ($A = U - TS$); Gibbs free energy ($G = H - TS$); criterion for spontaneity ($\Delta G_{T,P} < 0$); Nernst heat theorem and Third Law of Thermodynamics ($\lim_{T \to 0} S = 0$ for perfect crystals).
- **Gibbs-Helmholtz Equation:** Derivation of $\Delta G = \Delta H + T\left(\partial \frac{\Delta G}{\partial} T\right)_p$ and $\left[\partial \frac{\frac{\Delta G}{T}}{\partial} T\right]_p = -\Delta \frac{H}{T^2}$; temperature dependence of equilibrium constants (van 't Hoff isochore).
- **Chemical Kinetics & Rate Laws:** Reaction rate definitions; differential rate laws; reaction order vs molecularity; integrated rate equations and half-life periods for 0th, 1st, and 2nd order reactions; pseudo-first-order reactions.
- **Collision Theory & Catalysis:** Bimolecular collision theory, collision frequency (Z_{AB}), steric/orientation factor (P); Arrhenius equation ($k = Ae^{-\frac{E_a}{RT}}$); determination of activation energy (E_a); homogeneous and heterogeneous catalysis mechanisms.

1. Basic Concepts & The First Law of Thermodynamics

1.1 Thermodynamic Systems, Variables & States

- **System and Surroundings:** A thermodynamic system is the specific macroscopically macroscopic portion of the universe under physical or chemical investigation, delimited by real or imaginary boundaries:
  1. **Open System:** Can exchange both matter and thermal energy with its surroundings (e.g., an open reaction flask, a biological cell).
  2. **Closed System:** Can exchange energy (heat, work) but **not** matter with surroundings (e.g., a sealed chemical bomb calorimeter).
  3. **Isolated System:** Exchanges neither energy nor matter with surroundings ($dq = 0, dw = 0, dm = 0$) (e.g., a perfectly insulated thermos flask, the universe as a totality).
- **Properties of Systems:**
  - **Extensive Properties:** Depend directly on the quantity of matter present (Mass m, Volume V, Internal Energy U, Enthalpy H, Entropy S, Heat Capacity C).
  - **Intensive Properties:** Independent of the mass or extent of the system (Temperature T, Pressure P, Density ρ, Refractive Index n, Specific Heat c, Molar Enthalpy H_m).
- **State Functions vs Path Functions:**
  - **State Functions:** Thermodynamic variables whose values depend exclusively on the present equilibrium state of the system, completely independent of the historical path taken ($\Delta U = U_2 - U_1, \Delta H = H_2 - H_1, \Delta G = G_2 - G_1$). Mathematically, their differentials are exact ($\oint dX = 0$).
  - **Path Functions:** Quantities whose magnitudes depend on the specific path or mechanism followed during the transition between states (q heat, w work). Their differentials are inexact ($\delta q, \delta w$).

1.2 The First Law of Thermodynamics & Enthalpy

- **Mathematical Statement (Conservation of Energy):** Energy can neither be created nor destroyed; it can only be transformed from one form into another:

$$dU = \delta q + \delta w$$

where dU is the change in internal energy, δq is infinitesimal heat absorbed by the system ($+q$ for endothermic, $-q$ for exothermic), and δw is work done **on** the system by surroundings under IUPAC sign convention ($\delta w = -P_{\text{ext}} dV$).

- **Expansion Work (PV Work):**

$$w = -\int_{V_1}^{V_2} P_{\text{ext}} dV$$

- Free Expansion into Vacuum ($P_{\text{ext}} = 0$): $w = 0$.
- Reversible Isothermal Expansion of Ideal Gas ($P_{\text{ext}} = P_{\text{int}} = nR\frac{T}{V}$):

$$w_{\text{rev}} = -\int_{V_1}^{V_2} \frac{nRT}{V} dV = -nRT \ln\left(\frac{V_2}{V_1}\right) = -nRT \ln\left(\frac{P_1}{P_2}\right) = -2.303 nRT \log_{10}\left(\frac{V_2}{V_1}\right)$$

Because P_{int} is maintained marginally higher than P_{ext} throughout the expansion, **maximum work** is obtained during a reversible process.

- **Enthalpy (H) & Heat Capacities:** Enthalpy is defined as the heat content at constant pressure:

$$H = U + PV \quad \Longrightarrow \quad dH = dU + PdV + VdP$$

  At constant volume ($dV = 0$): $dq_v = dU = C_v dT$, where $C_v = \left(\partial \frac{U}{\partial} T\right)_v$. At constant pressure ($dP = 0$): $dq_p = dH = C_p dT$, where $C_p = \left(\partial \frac{H}{\partial} T\right)_p$.
- **Mayer's Relation Derivation:** For an ideal gas, $H = U + PV = U + nRT$. Differentiating with respect to temperature T:

$$\frac{dH}{dT} = \frac{dU}{dT} + nR \quad \Longrightarrow \quad C_p - C_v = nR \quad \left(C_{p,m} - C_{v,m} = R \approx 8.314 \text{ J/(mol K)}\right)$$

1.3 Reversible Adiabatic Expansion of an Ideal Gas

In an adiabatic process, the system is thermally insulated from surroundings ($\delta q = 0$):

$$dU = \delta w = -PdV$$

Since $dU = C_v dT$ and $P = \frac{RT}{V}$ for one mole:

$$C_v dT = -\frac{RT}{V} dV \quad \Longrightarrow \quad C_v \frac{dT}{T} = -R \frac{dV}{V}$$

Integrating between states (T_1, V_1) and (T_2, V_2) with $R = C_p - C_v$ and defining $\gamma = \frac{C_p}{C_v}$:

$$C_v \ln\left(\frac{T_2}{T_1}\right) = -(C_p - C_v) \ln\left(\frac{V_2}{V_1}\right) \quad \Longrightarrow \quad \ln\left(\frac{T_2}{T_1}\right) = (1-\gamma) \ln\left(\frac{V_2}{V_1}\right) = \ln\left(\left(\frac{V_1}{V_2}\right)^{\gamma-1}\right)$$

$$T_2 V_2^{\gamma-1} = T_1 V_1^{\gamma-1} \quad \Longrightarrow \quad TV^{\gamma-1} = \text{constant}$$

Substituting $T = \frac{PV}{R}$ yields the classic Poisson relation:

$$PV^{\gamma} = \text{constant}, \quad T^{\gamma} P^{1-\gamma} = \text{constant}$$

Work done in reversible adiabatic expansion:

$$w_{\text{adi}} = \Delta U = nC_v(T_2 - T_1) = \frac{nR(T_2 - T_1)}{\gamma - 1} = \frac{P_2 V_2 - P_1 V_1}{\gamma - 1}$$

2. The Second Law of Thermodynamics & The Carnot Cycle

2.1 Limitations of the First Law & Need for Second Law

The First Law asserts energy conservation but fails to dictate the **direction of spontaneous change** (e.g., heat flows spontaneously from hot to cold bodies, gas expands to fill a vacuum, but the reverse never occurs spontaneously without external work, despite conserving energy).

- **Second Law Statements:**

1. **Kelvin-Planck Statement:** It is impossible for any cyclic device to receive heat from a single thermal reservoir and deliver an equivalent amount of work without producing any other effect ($\eta < 100\%$; a 100% efficient heat engine is a physical impossibility).
2. **Clausius Statement:** It is impossible to construct a cyclic device whose sole effect is the transfer of heat from a cooler body to a hotter body without an input of external work (refrigerators require power).

2.2 The Carnot Cycle & Maximum Thermal Efficiency

Sadi Carnot (1824) devised an idealized theoretical engine operating on a closed reversible thermodynamic cycle using one mole of an ideal gas acting between a hot thermal source at temperature T_H and a cold sink at temperature T_C ($T_H > T_C$).

- **The Four Reversible Stages:**
  1. **Stage 1: Reversible Isothermal Expansion ($A \to B$ at T_H):** Gas absorbs heat q_H from the source at constant T_H, expanding from V_1 to V_2:
  $$\Delta U_1 = 0, \quad q_H = -w_1 = RT_H \ln\left(\frac{V_2}{V_1}\right)$$
  2. **Stage 2: Reversible Adiabatic Expansion ($B \to C, T_H \to T_C$):** Gas is insulated ($q = 0$), expanding from V_2 to V_3 while its temperature drops from T_H to T_C:
  $$w_2 = C_v(T_C - T_H) = -C_v(T_H - T_C)$$
  3. **Stage 3: Reversible Isothermal Compression ($C \to D$ at T_C):** Gas is placed on the cold sink at T_C and compressed from V_3 to V_4, rejecting heat q_C:
  $$\Delta U_3 = 0, \quad -q_C = w_3 = -RT_C \ln\left(\frac{V_3}{V_4}\right)$$
  4. **Stage 4: Reversible Adiabatic Compression ($D \to A, T_C \to T_H$):** Gas is thermally insulated and compressed from V_4 to V_1, restoring temperature to T_H:
  $$w_4 = C_v(T_H - T_C)$$
- **Total Net Work & Thermal Efficiency Derivation:** The adiabatic work terms cancel identically ($w_2 + w_4 = -C_v(T_H - T_C) + C_v(T_H - T_C) = 0$). Net work done per cycle:
  $$w_{\text{net}} = -w_1 - w_3 = RT_H \ln\left(\frac{V_2}{V_1}\right) - RT_C \ln\left(\frac{V_3}{V_4}\right)$$
  From adiabatic relationships for stages 2 and 4:
  $$T_H V_2^{\gamma-1} = T_C V_3^{\gamma-1} \quad \text{and} \quad T_H V_1^{\gamma-1} = T_C V_4^{\gamma-1}$$
  Dividing the two equations:
  $$\left(\frac{V_2}{V_1}\right)^{\gamma-1} = \left(\frac{V_3}{V_4}\right)^{\gamma-1} \implies \frac{V_2}{V_1} = \frac{V_3}{V_4}$$
  Substituting this volume equality into the work equation:
  $$w_{\text{net}} = R(T_H - T_C) \ln\left(\frac{V_2}{V_1}\right)$$
  Thermal efficiency η:
  $$\eta = \frac{w_{\text{net}}}{q_H} = \frac{R(T_H - T_C)\ln\left(\frac{V_2}{V_1}\right)}{RT_H \ln\left(\frac{V_2}{V_1}\right)} = \frac{T_H - T_C}{T_H} = 1 - \frac{T_C}{T_H}$$
- **Carnot's Theorems:**
  1. No heat engine operating between two given temperatures can be more efficient than a reversible Carnot engine operating between the same two temperatures.
  2. All reversible engines operating between the same two thermal reservoirs have identical efficiencies, completely independent of the working substance.

3. Entropy, Free Energy & The Gibbs-Helmholtz Equation

3.1 Concept of Entropy (S) & Third Law of Thermodynamics

From the Carnot cycle efficiency $\eta = 1 - \frac{q_C}{q_H} = 1 - \frac{T_C}{T_H}$:

$$\frac{q_H}{T_H} - \frac{q_C}{T_C} = 0 \implies \sum_k \frac{q_k}{T_k} = 0 \implies \oint \frac{dq_{\text{rev}}}{T} = 0$$

This line integral around a closed path vanishes, proving that $\frac{dq_{\text{rev}}}{T}$ is an exact differential of a state function termed **Entropy (S)**:

$$dS = \frac{dq_{\text{rev}}}{T}$$

- **Physical Interpretation:** Entropy is a quantitative measure of microscopic molecular disorder or randomness in a system (Boltzmann relation: $S = k_B \ln \Omega$).
- **Third Law of Thermodynamics (Nernst Heat Theorem):** The entropy of every pure, perfectly crystalline substance approaches zero as absolute temperature approaches zero Kelvin:

  $$\lim_{T \to 0\ \text{K}} S = 0$$

  This enables the determination of absolute standard entropies ($S°$) of chemical substances by numerical integration of heat capacity data: $S_T = \int_0^T \left(\frac{C_p}{T}\right) dT$.

3.2 Gibbs Free Energy & Derivation of Gibbs-Helmholtz Equation

Gibbs Free Energy is defined to describe spontaneity at constant temperature and pressure:

$$G = H - TS$$

- **Spontaneity Criteria at Constant T and P:**

  $$\Delta G = \Delta H - T\Delta S$$

  - If $\Delta G < 0$: Process is spontaneous (exergonic).
  - If $\Delta G = 0$: System is at dynamic thermodynamic equilibrium.
  - If $\Delta G > 0$: Process is non-spontaneous in the forward direction (endergonic).
- **Rigorous Derivation of Gibbs-Helmholtz Equation:** From fundamental thermodynamic relation: $dG = VdP - SdT$. At constant pressure ($dP = 0$):

  $$\left(\partial \frac{G}{\partial} T\right)_P = -S$$

  Substitute $S = -\left(\partial \frac{G}{\partial} T\right)_P$ into the definition $G = H - TS$:

  $$G = H + T\left(\partial \frac{G}{\partial} T\right)_P$$

  For an initial state 1 and final state 2 of a chemical reaction:

  $$G_1 = H_1 + T\left(\partial \frac{G_1}{\partial} T\right)_P \quad \text{and} \quad G_2 = H_2 + T\left(\partial \frac{G_2}{\partial} T\right)_P$$

  Subtracting state 1 from state 2 ($\Delta G = G_2 - G_1, \Delta H = H_2 - H_1$):

  $$\Delta G = \Delta H + T\left[\partial \frac{\Delta G}{\partial} T\right]_P$$

  Alternate Form (Differential of $\Delta \frac{G}{T}$): Differentiating $\left(\Delta \frac{G}{T}\right)$ with respect to T at constant P using quotient rule:

  $$\left[\frac{\partial}{\partial} T\left(\Delta \frac{G}{T}\right)\right]_P = \frac{T\left[\partial \frac{\Delta G}{\partial} T\right]_P - \Delta G}{T^2}$$

  From Gibbs-Helmholtz equation, $T\left[\partial \frac{\Delta G}{\partial} T\right]_P - \Delta G = -\Delta H$. Therefore:

$$\left[\frac{\partial}{\partial}T\left(\frac{\Delta G}{T}\right)\right]_P = -\frac{\Delta H}{T^2}$$

Since standard reaction free energy is linked to the equilibrium constant by $\Delta G^\circ = -RT\ln K_{\text{eq}}$, substituting yields the famous **van 't Hoff Isochore**:

$$\frac{d\ln K_{\text{eq}}}{dT} = \frac{\Delta H^\circ}{RT^2} \quad \Longrightarrow \quad \ln\left(\frac{K_2}{K_1}\right) = \frac{\Delta H^\circ}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

4. Chemical Kinetics: Rate Laws, Reaction Orders & Collision Theory

4.1 Reaction Rates, Order & Molecularity

For a generalized chemical reaction $aA + bB \rightarrow cC + dD$:

$$\text{Rate} = -\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = +\frac{1}{c}\frac{d[C]}{dt} = +\frac{1}{d}\frac{d[D]}{dt}$$

- **Rate Law:** An experimentally determined equation linking reaction rate to reactant concentrations:

$$\text{Rate} = k[A]^m[B]^n$$

where k is the specific reaction rate constant, and overall order is $n_{\text{total}} = m + n$.

- **Order vs Molecularity:**
  - **Order:** Experimental empirical quantity; sum of exponents in rate law; can be zero, fractional, or integer.
  - **Molecularity:** Theoretical concept; number of reacting species colliding simultaneously in an elementary step; strictly a positive whole integer $(1, 2, 3)$.

4.2 Integrated Rate Equations & Half-Life Derivations

Order	Differential Rate Law	Integrated Rate Equation	Half-Life ($t_{\frac{1}{2}}$)	Units of k
Zero	$-\frac{d[A]}{dt} = k$	$[A]_t = [A]_0 - kt$	$t_{\frac{1}{2}} = \frac{[A]_0}{2k}$	$\text{mol L}^{-1}\text{ s}^{-1}$
First	$-\frac{d[A]}{dt} = k[A]$	$k = \frac{1}{t}\ln\left(\frac{[A]_0}{[A]_t}\right) = \frac{2.303}{t}\log_{10}\left(\frac{[A]_0}{[A]_t}\right)$	$t_{\frac{1}{2}} = \frac{\ln 2}{k} = \frac{0.693}{k}$	s^{-1}
Second $(2A \rightarrow P)$	$-\frac{d[A]}{dt} = k[A]^2$	$\frac{1}{[A]_t} - \frac{1}{[A]_0} = kt$	$t_{\frac{1}{2}} = \frac{1}{k[A]_0}$	$\text{L mol}^{-1}\text{ s}^{-1}$

- **Derivation of First-Order Rate Equation:**

$$-\frac{d[A]}{[A]} = kdt \quad \Longrightarrow \quad \int_{[A]_0}^{[A]_t}\frac{d[A]}{[A]} = -k\int_0^t dt$$

$$\ln\left(\frac{[A]_t}{[A]_0}\right) = -kt \quad \Longrightarrow \quad [A]_t = [A]_0 e^{-kt}$$

Setting $[A]_t = \frac{[A]_0}{2}$ for half-life:

$$\ln\left(\frac{1}{2}\right) = -kt_{\frac{1}{2}} \quad \Longrightarrow \quad t_{\frac{1}{2}} = \frac{\ln 2}{k} = \frac{0.69315}{k}$$

Crucially, the half-life of a first-order reaction is **completely independent** of initial reactant concentration $[A]_0$.

4.3 Collision Theory & Arrhenius Equation

- **Bimolecular Collision Theory:** Reactant molecules must collide physically with proper orientation and sufficient kinetic energy equal to or exceeding the **Threshold Energy ($E_{\text{threshold}}$)**:

$$\text{Rate} = P \times Z_{AB} \times e^{-\frac{E_a}{RT}}$$

where Z_{AB} is collision frequency, P is steric (orientation) factor ($0 < P \leq 1$), and $e^{-\frac{E_a}{RT}}$ is the Maxwell-Boltzmann fraction of collisions possessing energy $\geq E_a$.

- **Arrhenius Equation:**

$$k = Ae^{-\frac{E_a}{RT}} \implies \ln k = \ln A - \frac{E_a}{RT} \implies \log_{10} k = \log_{10} A - \frac{E_a}{2.303RT}$$

where $A = PZ_{AB}$ is the pre-exponential frequency factor, and E_a is Activation Energy. Comparing two temperatures T_1 and T_2:

$$\ln\left(\frac{k_2}{k_1}\right) = \frac{E_a}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right] = \frac{E_a(T_2 - T_1)}{RT_1T_2}$$

- **Role of Catalysts:** A catalyst accelerates chemical reaction rates by providing an alternative reaction pathway with a lower activation energy ($E_{a'} < E_a$). It speeds up both forward and reverse reactions equally, leaving equilibrium constant K_{eq} and reaction free energy ΔG completely unchanged.

VIT FAT Solved Question Bank: Step-by-Step Numerical Derivations

FAT Problem 1: Carnot Cycle Heat Engine Efficiency & Reversible Gas Work

Question (10 Marks): A Carnot heat engine operates between a source at $T_H = 600°\text{C}$ and a sink at $T_C = 30°\text{C}$. In each cycle, the engine absorbs 2500 kJ of heat from the high-temperature reservoir. ("i") Calculate the theoretical maximum thermal efficiency η of this engine. ("ii") Determine the mechanical work output w_{net} produced per cycle in kilojoules. ("iii") Calculate the amount of heat q_C discharged to the low-temperature reservoir per cycle. ("iv") If the same engine is operated in reverse as a Carnot refrigerator between the same two temperatures, determine its Coefficient of Performance (COP_{ref}).

Complete Step-by-Step Solution:

Step 1: Convert Temperatures to Absolute Scale (Kelvin):

$$T_H = 600 + 273.15 = 873.15 \text{ K}$$

$$T_C = 30 + 273.15 = 303.15 \text{ K}$$

Step 2: Calculate Thermal Efficiency η:

$$\eta = \frac{T_H - T_C}{T_H} = \frac{873.15 - 303.15}{873.15} = \frac{570}{873.15} = 0.6528 \quad (65.28\%)$$

Step 3: Mechanical Work Output w_{net} Per Cycle:

$$w_{\text{net}} = \eta \times q_H = 0.6528 \times 2500 \text{ kJ} = 1632.0 \text{ kJ}$$

Step 4: Heat Rejected to Cold Reservoir q_C: From energy conservation ($w_{\text{net}} = q_H - q_C$):

$$q_C = q_H - w_{\text{net}} = 2500 - 1632.0 = 868.0 \text{ kJ}$$

Verification: $q_C = q_H\left(\frac{T_C}{T_H}\right) = 2500 \times \left(\frac{303.15}{873.15}\right) = 2500 \times 0.3472 = 868.0$ kJ. Matches perfectly.

Step 5: Coefficient of Performance as Refrigerator:

$$\text{COP}_{\text{ref}} = \frac{T_C}{T_H - T_C} = \frac{303.15}{873.15 - 303.15} = \frac{303.15}{570} = 0.5318$$

Boxed Final Answers:

$$\text{(i)}\eta = 65.28\%, \quad \text{(ii)}w_{\text{net}} = 1632.0 \text{ kJ}, \quad \text{(iii)}q_C = 868.0 \text{ kJ}, \quad \text{(iv)}\text{COP}_{\text{ref}} = 0.532.$$

FAT Problem 2: Gibbs-Helmholtz Equation & Reaction Spontaneity

Question (10 Marks): For a chemical synthesis reaction at standard atmospheric pressure, the change in free energy is found to be $\Delta G = -120.5$ kJ at 300 K and -115.2 kJ at 310 K. ("i") Calculate the temperature coefficient of free energy $\left[\partial \frac{\Delta G}{\partial} T\right]_P$ assuming linear variation over this temperature range. ("ii") Using the Gibbs-Helmholtz equation, determine the reaction enthalpy ΔH at 300 K. ("iii") Calculate the standard entropy change of the reaction ΔS at 300 K. ("iv") Comment on whether the reaction is endothermic or exothermic, and determine if it is enthalpy-driven or entropy-driven.

Complete Step-by-Step Solution:

Step 1: Calculate Temperature Coefficient of Free Energy:

$$\left[\partial \frac{\Delta G}{\partial} T\right]_P \approx \frac{\Delta G_2 - \Delta G_1}{T_2 - T_1} = \frac{-115.2 - (-120.5)}{310 - 300} = \frac{+5.3 \text{ kJ}}{10 \text{ K}} = +0.53 \text{ kJ/K} = +530 \text{ J/K}$$

Step 2: Determine Reaction Enthalpy ΔH using Gibbs-Helmholtz: The Gibbs-Helmholtz equation states:

$$\Delta G = \Delta H + T\left[\partial \frac{\Delta G}{\partial} T\right]_P$$

Rearranging for enthalpy ΔH:

$$\Delta H = \Delta G - T\left[\partial \frac{\Delta G}{\partial} T\right]_P$$

Substitute $T = 300$ K, $\Delta G = -120.5$ kJ, and $\left[\partial \frac{\Delta G}{\partial} T\right]_P = +0.53$ kJ/K:

$$\Delta H = -120.5 \text{ kJ} - (300 \text{ K} \times 0.53 \text{ kJ/K}) = -120.5 - 159.0 = -279.5 \text{ kJ}$$

Step 3: Calculate Entropy Change ΔS: Since $\left[\partial \frac{\Delta G}{\partial} T\right]_P = -\Delta S$:

$$\Delta S = -\left[\partial \frac{\Delta G}{\partial} T\right]_P = -0.53 \text{ kJ/K} = -530 \text{ J/K}$$

Verification using $\Delta G = \Delta H - T\Delta S$:

$$\Delta G = -279.5 - (300 \times (-0.53)) = -279.5 + 159.0 = -120.5 \text{ kJ}$$

. Verified.

Step 4: Spontaneity & Thermodynamic Driving Force:

- Since $\Delta H = -279.5$ kJ < 0, the reaction is strongly **exothermic**.
- Since $\Delta S = -530$ J/K < 0, disorder decreases (disfavored by entropy).
- The reaction is strictly **enthalpy-driven**; the large negative enthalpy term (-279.5 kJ) overcomes the opposing positive entropy penalty ($-T\Delta S = +159.0$ kJ), driving ΔG negative.

Boxed Final Answers:

$$\left[\partial \frac{\Delta G}{\partial} T\right]_P = +0.53 \text{ kJ/K}, \quad \Delta H = -279.5 \text{ kJ}, \quad \Delta S = -530 \text{ J/K}, \quad \text{Strongly exothermic \& enthalpy-driven.}$$

FAT Problem 3: Arrhenius Activation Energy & First-Order Half-Life Kinetics

Question (10 Marks): The rate constant for the thermal decomposition of a chemical propellant is $k_1 = 3.5 \times 10^{-4}$ s^{-1} at 300 K and $k_2 = 3.15 \times 10^{-2}$ s^{-1} at 350 K. ("i") Determine the activation energy E_a for the decomposition reaction in kJ/mol. ("ii") Calculate the Arrhenius frequency factor A. ("iii") Determine the rate constant k and half-life $t_{\frac{1}{2}}$ at an operating temperature of 325 K. ("iv") Calculate the time required for 90% decomposition of the propellant at 325 K.

Complete Step-by-Step Solution:

Step 1: Calculate Activation Energy E_a: The integrated Arrhenius equation is:

$$\ln\left(\frac{k_2}{k_1}\right) = \frac{E_a}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right] = \frac{E_a(T_2 - T_1)}{RT_1T_2}$$

$$\frac{k_2}{k_1} = \frac{3.15 \times 10^{-2}}{3.5 \times 10^{-4}} = 90.0$$

$$\ln(90.0) = 4.4998$$

$$\frac{T_2 - T_1}{T_1T_2} = \frac{350 - 300}{300 \times 350} = \frac{50}{105000} = 4.7619 \times 10^{-4}\ \text{K}^{-1}$$

With $R = 8.314$ J/(mol K):

$$4.4998 = \frac{E_a \times 4.7619 \times 10^{-4}}{8.314}$$

$$E_a = \frac{4.4998 \times 8.314}{4.7619 \times 10^{-4}} = \frac{37.411}{4.7619 \times 10^{-4}} = 78563\ \text{J/mol} = 78.563\ \text{kJ/mol}$$

Step 2: Determine Arrhenius Frequency Factor A: From $k_1 = Ae^{-\frac{E_a}{RT_1}}$:

$$\ln A = \ln k_1 + \frac{E_a}{RT_1} = \ln(3.5 \times 10^{-4}) + \frac{78563}{8.314 \times 300} = -7.9576 + \frac{78563}{2494.2} = -7.9576 + 31.498 = 23.5404$$

$$A = e^{23.5404} = 1.673 \times 10^{10}\ \text{s}^{-1}$$

Step 3: Rate Constant & Half-Life at 325 K (52°C):

$$\ln\left(\frac{k_3}{k_1}\right) = \frac{E_a}{R}\left[\frac{1}{T_1} - \frac{1}{T_3}\right] = \frac{78563}{8.314}\left[\frac{1}{300} - \frac{1}{325}\right] = 9449.5 \times \left[\frac{325 - 300}{300 \times 325}\right] = 9449.5 \times \left(\frac{25}{97500}\right)$$

$$\ln\left(\frac{k_3}{k_1}\right) = 9449.5 \times 2.5641 \times 10^{-4} = 2.423$$

$$\frac{k_3}{k_1} = e^{2.423} = 11.279$$

$$k_3 = 11.279 \times (3.5 \times 10^{-4}) = 3.948 \times 10^{-3}\ \text{s}^{-1}$$

Half-life at 325 K:

$$t_{\frac{1}{2}} = \frac{0.69315}{k_3} = \frac{0.69315}{3.948 \times 10^{-3}} = 175.57\ \text{seconds} \approx 2.93\ \text{minutes}$$

Step 4: Time for 90% Decomposition: For 90% reaction, $[A]_t = 0.10[A]_0$:

$$t_{90\%} = \frac{1}{k_3}\ln\left(\frac{[A]_0}{0.10[A]_0}\right) = \frac{\ln 10}{k_3} = \frac{2.3026}{3.948 \times 10^{-3}} = 583.23\ \text{seconds} \approx 9.72\ \text{minutes}$$

Boxed Final Answers:

$$E_a = 78.56\ \text{kJ/mol}, \quad A = 1.67 \times 10^{10}\ \text{s}^{-1}, \quad k_{325\text{K}} = 3.95 \times 10^{-3}\ \text{s}^{-1}, \quad t_{\frac{1}{2}} = 175.6\ \text{s}, \quad t_{90\%} = 583.2\ \text{s}.$$